



AJKD Atlas of Renal Pathology: Type III Collagen Glomerulopathy

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Clinical and Pathologic Features

Type III collagen glomerulopathy (also called collagenofibrotic glomerulopathy) may occur at any age, with males and females affected equally. Childhood onset with familial occurrences has suggested autosomal recessive inheritance in some, while affected adults appear to represent sporadic cases. Patients present with proteinuria (often nephrotic-range), hypertension, and subsequent slowly progressive chronic kidney disease, with many reaching end-stage kidney disease within 10 years. Very rare cases of extrarenal involvement have been described. In some patients, there is associated hemolytic uremic syndrome, with corresponding clinical manifestations and more rapid decline in kidney function.

Light microscopy: Glomeruli show mesangial expansion with normal or mild increase in mesangial cellularity, and marked mesangial expansion with weakly periodic acid–Schiff–positive material. Glomerular basement membranes (GBMs) show a double contour, and glomeruli may appear nodular or lobular in more advanced cases. There is proportional tubulointerstitial fibrosis and nonspecific vascular sclerosis.

Immunofluorescence microscopy: There are no immune complexes by standard immunofluorescence microscopy. Stains for type III collagen show strong signal within the mesangium and along GBMs.

Electron microscopy: Massive type III collagen banded fibrils with about 60 nm periodicity are visualized when tissue has been stained with phosphotungstic acid, expanding mesangial and sub-endothelial areas, but not permeating the lamina densa of the GBM. These fibrils are randomly arranged and curved, contrasting the normal straight appearance of

type III collagen. Overlying podocyte foot processes are effaced.

Etiology/Pathogenesis

The etiology and pathogenesis are unknown. The familial occurrence in some young patients suggests autosomal recessive inheritance. A canine model captures many of the elements of the human disease and also shows autosomal recessive inheritance. No mutation has been found in type III collagen in the animal model or in affected humans. Rare patients with factor H deficiency and hemolytic uremic syndrome and type III collagen glomerulopathy have been described. However, the mechanism for the coexistence of these 2 disease manifestations is not known. Patients with type III collagen glomerulopathy show increased precursor of procollagen type III in circulation, but this is not a specific or unique marker of the disease.

Differential Diagnosis

Fibrillar collagen may be seen nonspecifically in any chronic scarring disease. Collagen fibrils may be more prominent than usual in some cases of diabetic nephropathy, a condition called diabetic fibrillosis. In type III collagen glomerulopathy, the collagen type III deposits are massive. In contrast to nail-patella syndrome, the lamina densa is spared, with type III collagen only present within the mesangium and in subendothelial areas. Another rare entity, fibronectin glomerulopathy, also can have organized fibrillar deposits, but these lack the banded character of type III collagen. The diagnosis of fibronectin glomerulopathy can be further established by immunohistochemical stains for fibronectin. Other causes of a membranoproliferative pattern, such as immune-complex diseases or monoclonal immunoglobulin deposition disease, are distinguished by specific immunofluorescence positivity. Amyloid is distinguished by Congo Red positivity, with straight, small, non-cross-banded amyloid fibrils.

Key Diagnostic Features

- Lobular pattern with membranoproliferative appearance with double contours of GBM
- No immune complexes by immunofluorescence
- Massive banded fibrillar type III collagen in mesangium and subendothelial area

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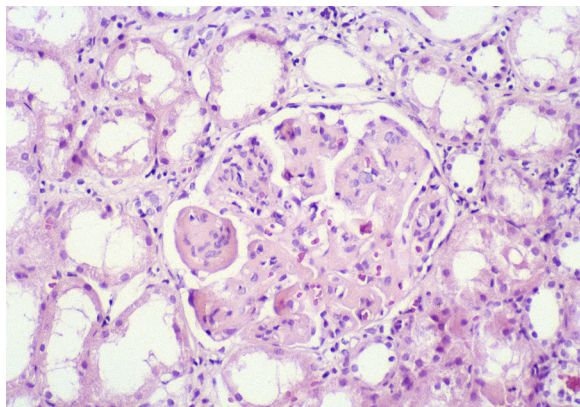


Figure 1. Type III collagen glomerulopathy shows enlarged lobular glomeruli with variable mesangial hypercellularity and pale amorphous expansion of mesangial matrix (hematoxylin and eosin stain). Reproduced with permission from *AJKD* 38(4):e15-e16.

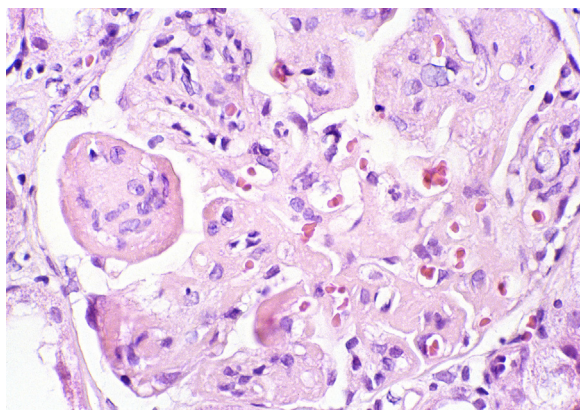


Figure 2. Type III collagen glomerulopathy shows enlarged lobular glomeruli with variable mesangial hypercellularity and pale amorphous expansion of mesangial matrix that fills capillary loops (hematoxylin and eosin stain). Reproduced with permission from *AJKD* 38(4):e15-e16.

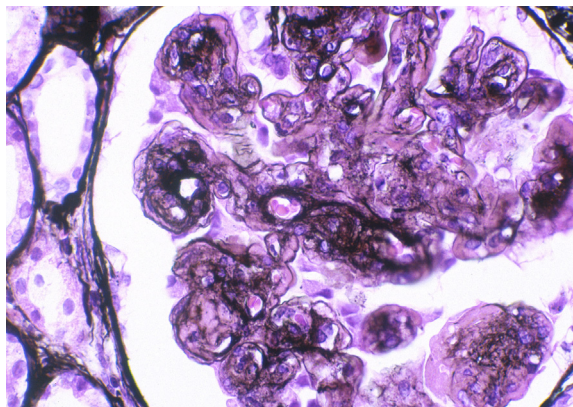


Figure 3. Type III collagen glomerulopathy with membranoproliferative pattern of injury (Jones silver stain). Standard immunofluorescence studies (not shown) display no staining. Reproduced with permission from *AJKD* 38(4):e15-e16.

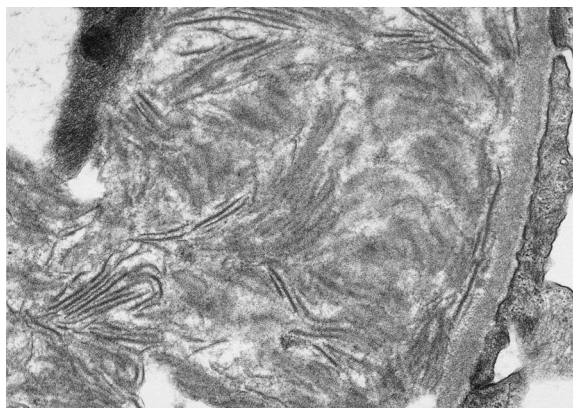


Figure 4. Type III collagen glomerulopathy with large abundance of fibrillary type III collagen subendothelial deposits with whorling and cross-striated appearance (electron microscopy). Reproduced with permission from *AJKD* 38(4):e15-e16.